

Modeling and analysis of business process reengineering strategies for improving emergency department efficiency

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Abstract

Emergency departments (ED) in the USA treat 136.9 million cases annually and account for nearly half of all medical care delivered. Due to high demand and limited resource (such as doctors and beds) availability, the median waiting time for ED patients is around 90 minutes. This research is motivated by a real-life case study of an ED located in central Missouri, USA, which faces the problem of congestion and improper workload distribution (e.g., overburdened ED doctors). The objective of this paper is to minimize patient waiting time and efficiently allocate workload among resources in an economical manner. The systematic framework of Business Process Reengineering (BPR), along with discrete-event simulation modeling approach, is employed to analyze current operations and potential improvement strategies. Alternative scenarios pertaining to process change, workforce planning, and capacity expansion are proposed. Besides process performance measures (waiting time and resource utilization), other criteria, such as responsiveness, cost of adoption, and associated risk, are also considered for evaluating an alternative. The experimental analysis indicates that a change in the triage process (evenly distributing medium-acuity patients among doctors and mid-level providers) is economical, easy to implement, reduces physician workload, and improves average waiting time by 20%, thereby making it attractive for short-term adoption. On the other hand, optimizing the workforce level based on historical demand patterns while adopting a triage process change delivers the best performance (84% reduction in waiting time and balanced resource utilization), and is recommended as a long-term solution.

Keywords

Emergency department, waiting time, workforce planning, business process reengineering, discrete-event simulation, process change

1. Introduction

Emergency Department (ED) overcrowding, a situation in which the number of patients waiting for service/treatment far exceeds the available capacity, is a significant concern across the globe.¹ Specifically, ED overcrowding is prevalent in the US as it accounts for nearly half of all the medical care delivered.² The demand for emergency care has increased by 50% in the last 20 years, but the number of ED units has declined by 11%.³ This rising demand can be attributed to the aging population, increase in chronic medical conditions, uninsured patients, and non-urgent visits. Consequently, it has detrimental effects on the patient and hospital staff. Besides long waiting times, it may result in patient dissatisfaction and improper care, subsequently leading to increased risk of mortality, readmission, and

medical errors.⁴ For instance, the mortality rate at an overcrowded ED is found to be 5% higher than a less crowded ED.⁵ As the ED demand in the US increased by 59% from 1999 to 2009, the readmission rates also increased by 36%.⁶ Likewise, the physical and mental well-being of the ED staff is also affected, which could later lead to burnout

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and suicide. It is estimated that over 60% of the emergency physicians experience burnout—the highest rate among all specialties. Moreover, the physician suicide rate is substantially higher than the general population—70% for men and 250–400% for women.⁷ Therefore, it is imperative to mitigate ED crowding and reduce physician workload by streamlining the care process.

The typical ED process in the US involves five stages: registration, triage, examination, treatment, and discharge or transfer process.⁸ An arriving patient must visit each stage in the process, except when the patient is in an extremely critical state (e.g., injured and unresponsive, cardiac arrest, anaphylactic shock). The patient first checks in at the registration desk, followed by triage, where he/she is classified based upon his/her acuity level. The triage nurses follow a hospital established protocol (e.g., Emergency Severity Index) to categorize the patients into different acuity levels. Critical patients are given priority in boarding and treatment processes (nurse assessment and medical diagnosis/treatment by a physician assistant or doctor). The allocation of resources to the patients is also decided based on the acuity level of the patients. Generally, a physician assistant is assigned for a low-acuity case, whereas a doctor attends a patient with high acuity. However, such practice might be uncommon outside the US.

The cause of crowding is identified at different stages of an emergency care process, namely, arrival, treatment, and discharge.⁹ For instance, increasing ED visits (patient demand) is one of the critical reasons for ED crowding, and is attributed to the arrival stage.¹⁰ Some of the notable contributors to the surge in ED demand, such as aging and growing populations, are difficult to avoid or manage at the operational level.¹¹ On the other hand, other important factors, such as the high volume of less severe cases visiting the ED, can be controlled to alleviate the rising demand.¹² At the treatment stage, shortage of resources (nurse, resident assistant, doctors, etc.) and delayed test results (magnetic resonance imaging (MRI), interventional radiology (IR), ultrasound, X-ray, computed tomography (CT) scan) become the bottleneck, resulting in lower patient throughput, which then contributes to congestion within the ED.¹⁰ Finally, at the discharge stage, the inability to quickly process patients from ED to inpatient beds or home is identified as the primary cause for blocking access to the arriving patients.¹³

This research is motivated by a real-life case study of an ED located in rural Missouri which experiences overcrowding, with an average waiting time of 87 minutes. Moreover, ED doctors are overburdened because of the continuous influx of patients. To address these adverse effects of ED crowding and improve service quality, this paper seeks to minimize patient wait times and reduce the provider's workload in an economical manner.

The remainder of this paper is structured as follows. Section 2 presents a brief review of some of the notable

works on ED crowding. Section 3 discusses the problem description and research methodology. The experimental results and discussion are presented in Section 4. Finally, conclusions and directions for future work are outlined in Section 5.

2. Literature review

Extensive research has been done on mitigating the impact of ED crowding. Most of the previous work aimed to address the issues in one or more stages of the emergency care process—arrival, treatment, and discharge. In this section, some of the prominent works addressing the challenges arising in these three stages are reviewed.

Previous research on tackling increasing ED demand (arrival) has adopted different strategies to dissuade non-urgent arrivals. Sharma and Inder¹⁴ studied the impact of directing the low-acuity patients visiting EDs to a co-located outpatient facility using a survival regression technique and observed a 19% reduction in ED waiting time. Another approach for reducing undesirable ED visits was to increase the working hours of primary care centers.¹⁵ Whittaker et al.¹⁶ investigated the effect of extended operating hours of a co-located outpatient clinic and reported a 26.4% reduction in ED visits for minor problems. Likewise, Dolton and Pathania¹⁷ conducted a pilot study to assess the impact of increasing access to a nearby primary care clinic from 5 days to 7 days a week. Their initiative reduced ED attendance on weekdays and weekends by 9.9% and 17.9%, respectively. Alternatively, Anantharaman¹⁸ studied various social interventions on ED overcrowding and concluded that public education and financial disincentives could minimize unnecessary ED attendances.

Although diverting non-urgent visits has shown promising results in managing ED overcrowding, it may not be feasible or easy to adopt in many cases. Therefore, a majority of the studies focused on expediting the patient flow within the system, as those factors are relatively easy to control.^{19–21} The potential solutions included “front-end” care provided by the physician before boarding¹⁹ and physician-led triage (using a physician for the triage process instead of a nurse) to expedite throughput of non-critical patients.²² Sullivan et al.²³ analyzed the effect of process reforms (e.g., bed management, physician training, and cross-disciplinary collaboration) on ED access at an underperforming tertiary care hospital. The authors conducted a pre-post analysis of the interventions (or process reforms) using retrospective data and identified a significant improvement in outcome measures such as length of stay (LOS), discharge rate, and mortality rate. Han et al.²⁴ investigated physician-led triage and observed a 11-minute reduction in ED length of stay (EDLOS) for non-admitted patients (patients who are not transferred to inpatient unit

after visiting ED), while it did not change significantly for admitted patients. Categorizing patients based upon the acuity level and treating non-critical patients in a separate area also improved patient throughput and decreased ED crowding.²⁵ Bal et al.²⁶ improved patient throughput by using a value stream map to identify the bottlenecks and non-value added activities within the system. Discrete-event simulation (DES) was adopted to evaluate different alternative scenarios, such as process change and capacity expansion. The authors reported a 34% decrease in patient turnaround time, and concluded that a combination of lean techniques and simulation modeling would be an effective approach for improving patient turnaround time. Time spent during laboratory tests also plays a vital role in improving ED crowding, as previous research reported a significant reduction in LOS by using point-of-care testing (POCT).²⁷ Jang et al.²⁸ made a comparison between the POCT and the traditional central laboratory system by studying 10,224 randomly selected non-critical patients and concluded that POCT reduced LOS by 22 minutes. In another similar study, Begaz et al.²⁹ investigated the effect of initiating laboratory tests in the ED waiting room and observed a reduction of 44 minutes in LOS.

The introduction of an ED flow coordinator (senior nurse) stimulated improvements in patient flow within the ED and expedited the throughput.³⁰ Tenbensen et al.³¹ investigated the impact of introducing a nurse flow coordinator (a senior nurse who is responsible for communication and coordination of all ED activities from admission to discharge) at four ED hospitals and reported a substantial reduction in ED crowding. Mumma et al.³² studied the effect of increasing the number of ED beds from 33 to 53 on service quality. The authors found no significant improvement in the number of patients who were leaving without being treated. Furthermore, the augmented capacity had an unintended outcome of an increase in ED boarding time.

Finally, some research has substantially lowered ED crowding by reducing the ED discharge wait time. Burley et al.³³ evaluated the impact of giving ED patients a higher priority for the inpatient beds and reported a significant decrease in the median EDLOS. Barrett et al.³⁴ achieved a 21% reduction in the ED discharge wait time by introducing a bed management strategy that established a post-procedure admission (e.g., priority-admission, temporary placement areas) to inpatient units. Patel et al.³⁵ evaluated a leadership-based program to expedite the inpatient admission process of the ED patients by using a real-time system to track the bed status in the ED and inpatient unit. The authors reported a significant decrease in EDLOS and an increase in patient satisfaction. Table 1 summarizes the characteristics of some of the notable works on reducing ED overcrowding and compares it with this study.

In summary, most of the improvement strategies could be categorized as a business process reengineering (BPR) effort—an approach for revamping the core business process to improve system performance. The inherent steps in BPR, such as examining the current process and assessing the value of potential alternatives, can be conducted using simulation models in an economical and timely manner.^{36,37} Specifically, prior studies have used simulation to successfully perform the BPR steps for addressing ED overcrowding.³⁸ Shim and Kumar³⁹ conducted a BPR study of emergency care by evaluating the impact of current and expanded capacity levels. They concluded that patient waiting time could be reduced by 41% by adding an extra payment station. In another BPR study, Vasiliki et al.⁴⁰ improved LOS and waiting time at two ED by examining current operations and evaluating alternative patient flow strategies. Besides addressing ED overcrowding, simulation-based BPR study has also been adopted to improve the efficiency of surgical ward and registration process at hospitals.^{41,42}

This paper aims to alleviate the problem of ED overcrowding and deviates from the traditional literature in the following ways. First, the impact on ED performance measures when physician extenders (i.e., mid-level providers such as physician assistants) are used for serving a fraction of intermediate acuity cases (level 3 patients) is assessed. Typically, physician assistants treat low-acuity cases (levels 4 and 5), but the current guidelines do not restrict their service for intermediate or higher acuity cases.⁴³ As a result, physician assistants are assigned to serve moderate acuity cases on an ad-hoc basis at certain EDs depending on the state law and hospital policy.^{44,45} However, to the best of the authors' knowledge, the impact of such a practice on patient waiting time and resource utilization has not been extensively examined by most prior works (as indicated in Table 1). Second, a comprehensive list of factors (or criteria) for evaluating alternative scenarios and aiding decision-makers is adopted. Almost all the previous studies determine the best alternative based on one or more process performance metrics (e.g., waiting time, throughput), as shown in Table 1. However, real-life applications involve several other factors (such as cost and ease of implementing the alternative), and the trade-offs between them could affect the stakeholder's decision. Finally, a case study with actual data obtained from an ED in mid-Missouri is used to demonstrate the simulation-based process reengineering approach for analyzing the current setting and potential improvement strategies.

3. Methodology

In this section, a systematic approach to address the problem of long waiting times and inefficient resource

Table I. Summary of some prominent research on improving ED Efficiency.

Author	Objective		Improvement Strategy							Criteria for Evaluating Alternatives								
	ED Visits	Length of Stay	Resource Utilization	Turnaround Time	Waiting Time	Capacity Management	Diverting low-acuity cases	Expedite Discharge Process	Extend Hours of Co-located Outpatient Clinic	Patient Flow Management	Point-of-care Testing	Workforce Allocation	Split level 3 cases between physician assistants and doctors	Ease of Adoption	Implementation Cost	Performance	Responsiveness	Risk
Abo-Hamad and Arisha ⁵³	✓					✓				✓		✓				✓		
Bal et al. ²⁶				✓		✓										✓		
Barrett et al. ³⁴	✓							✓								✓		
Begaz et al. ²⁹	✓										✓					✓		
Brenner et al. ⁵⁴			✓			✓										✓		
Burley et al. ³³	✓				✓			✓								✓		
Dolton and Pathania ¹⁷	✓								✓							✓		
Haghighinejad et al. ⁵⁵					✓											✓		
Han et al. ²⁴	✓															✓		
Holroyd et al. ²²	✓									✓						✓		
Kumar and Shim ³⁸					✓	✓				✓						✓		
Jang et al. ²⁸	✓										✓					✓		
Sharma and Inder ¹⁴					✓		✓					✓				✓		
Sinreich and Jabali ⁵⁶	✓		✓			✓										✓		
Sullivan et al. ²³	✓					✓										✓		
Tenbelsel et al. ³⁰				✓						✓						✓		
Whittaker et al. ¹⁶	✓		✓		✓	✓			✓			✓	✓	✓	✓	✓	✓	✓
Current Study															✓			

allocation in ED is presented. Specifically, the following five-step simulation-integrated BPR procedure is conducted by adapting the methodology frameworks presented in the literature.^{36,46–48}

- *Defining the ED process:* The first step of the BPR procedure involves system observation, comprehending patients' feedback, and interaction with key stakeholders for problem identification and project goals formulation.
- *Understanding existing ED process:* This step deals with process mapping and collection of raw historical data pertaining to the system under study using existing sources, such as electronic medical records (EMRs) and other methods such as time study, site visits, and focus groups/interviews. It also entails the use of charts and descriptive statistics for understanding the current system measures such as average arrivals per hour, average patient waiting time, and average utilization of resources (nurse, doctor, physician assistants, and beds).
- *Analyzing current (or as-is) ED operations:* In the analysis of the as-is process, the data collected are fit to a theoretical distribution, which in turn, is used as inputs to develop a simulation model of the current operation. This is a critical step that identifies the core issues needing to be resolved.
- *Designing to-be (alternative or reengineered) process:* This step focuses on the identification of alternative strategies to enhance the performance of the current system as well as the design and development of these scenarios as a simulation model.
- *Test and evaluate alternative process:* Finally, the performance of the baseline model is compared with alternative scenarios. The alternative processes must be evaluated with respect to different criteria such as cost and ease implementation, as this provides insight for managerial recommendations.

4. Case study

The simulation-integrated BPR procedure (discussed in Section 3) is data-driven and requires site visits, a thorough understanding of the current process, and collectible and measurable data. Therefore, to illustrate the proposed approach, a real-life case study of a teaching hospital ED located in mid-Missouri is considered.

4.1. System observation and process description

The ED under study is located in rural Missouri and has an average annual demand of 35,000 patients. It has the only level 1 trauma center in central Missouri and serves a population that covers 18 counties. The ED operates for 24

hours and 7 days a week with two shifts (Shift 1: 8 am–8 pm, Shift 2: 8 pm–8 am). For each shift, the ED employs a total of 10 healthcare practitioners (two physicians, seven nurses, one physician assistant). Further, the ED has four trauma beds (reserved for very critical patients) and 23 general beds.

The authors interacted with stakeholders such as nurses, doctors, and hospital administrators to understand their issues. The prevailing concern among the stakeholders was ED overcrowding. Moreover, hospital administrators were willing to increase the bed capacity to alleviate this issue. However, the ED physicians perceived their workload to be overburdening and emotionally exhaustive over the last few months. To understand the patients' perception of ED service quality, the satisfaction surveys completed by over 5000 discharged patients were analyzed. Patients provide their rating (on a scale of 1–poor to 5–excellent) for different aspects of their ED visit: process (e.g., registration efficiency, wait time), healthcare provider (e.g., willingness to listen, time spent with patient), and facility (e.g., signage, comfort, cleanliness, adequate parking). Although a majority of patients provided a high rating for most aspects, over 67% were unhappy with the wait time. Therefore, the ED under study faces the problem of overcrowding leading to inordinate wait time and overburdened doctors. To improve this situation, the current process flow was first analyzed. Figure 1 illustrates the typical patient flow in the ED under study.

A patient may arrive at the ED either by ambulance or walk-in. A patient who walks-in is required to complete the registration and proceed to triage, where a nurse would classify the patient based on the acuity from level 1–5 (level 1 being the most critical and level 5 being non-urgent). Critical patients (level 1) are given priority over all others and never have to wait for any resources. Similarly, the preference of the patients waiting for resources is based upon the acuity level, with more critical patients being preferred above the less critical ones; a patient arriving by ground or air ambulance is categorized as level 1 by default, thereby bypassing the registration and triage process.

After the boarding process is complete, the patient is first assessed by a nurse followed by the doctor or physician assistant. Doctors only assess patients with acuity levels 1–3, whereas physician assistants are responsible for the assessment of the patients with acuity levels 4 and 5. Once the evaluation by a doctor or physician assistant is complete, the patient is either discharged after treatment or prescribed one or more radiology tests (X-ray, CT scan, IR, MRI, ultrasound). If an additional procedure is recommended, then the patient is scheduled for a same-day examination. Upon obtaining the test results, the doctor or physician assistant reviews it and recommends discharge or inpatient transfer accordingly.

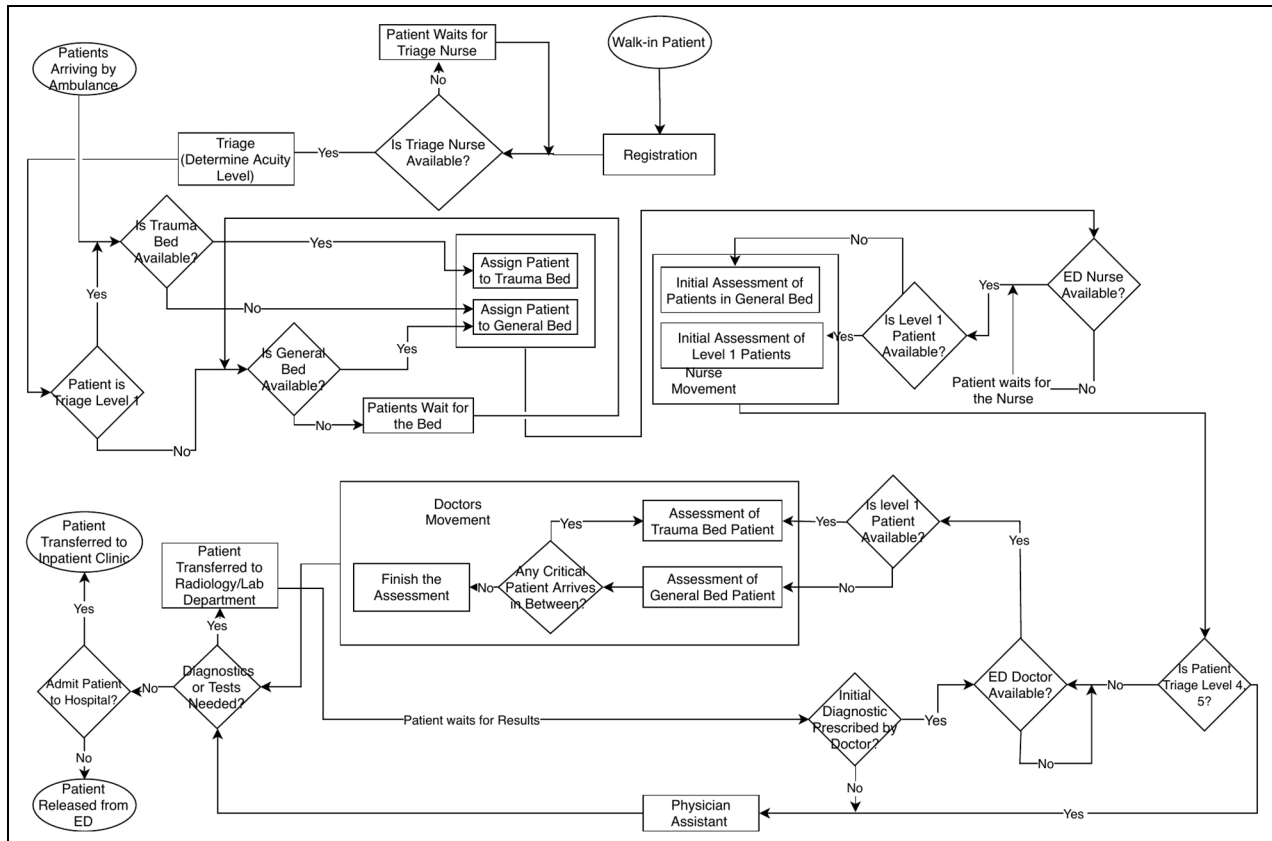


Figure 1. The process flow of the patient in the emergency department.

4.2. Data collection, analysis, and visualization

The data pertaining to the ED were obtained from the hospital's EMRs and included 35,000 patient visits between October 2017 and September 2018. For each patient visit, the extracted data included the following information:

- patient's arrival mode (ambulance or walk-in)
- arrival time
- triage level (levels 1–5)
- doctor assessment begin time and end time
- tests prescribed (e.g., X-ray, MRI)
- test begin and end time
- discharge time of patients

Of the 35,000 visits, 59% walked-in to ED, and 41% of the patients arrived by ambulance. Figure 2 shows the average number of ED arrivals on an hourly basis throughout the day, which was also used as the parameter for the non-homogenous Poisson process. Among the total number of patients that arrived as a walk-in, a substantial proportion (60%) were categorized as level 3, while 20% were classified as level 2, 0.41% as level 1, and 19.13% were

classified as either level 4 or 5. Also, over 70% of the patients arriving at the ED required radiology tests. The average time taken from patient arrival to discharge was 247 minutes, and the average door-to-provider time for a patient was 87 minutes. As the EMR data did not include the duration for registration and triage, a time study was conducted to estimate their values. To account for variability in patient flow and staff efficiency throughout the day, the time-study data were collected for 7 days during the two shifts. On average, 25 patients were observed in each shift, resulting in a total of 525 recordings. The average time taken by the registration clerk and triage nurse was estimated as 3 (min: 2 and max: 5) minutes and 10 (min: 5 and max: 15) minutes, respectively. The service time statistics were also verified by the experienced clinical staff members who are responsible for these processes on a daily basis.

Typical of any real-world database, the EMR data had missing values, inconsistent records (e.g., test end time being earlier than begin time), and outliers, which were discarded during the data pre-processing phase. The cleansed data were then used to determine the theoretical distribution for patient arrival time and resource service

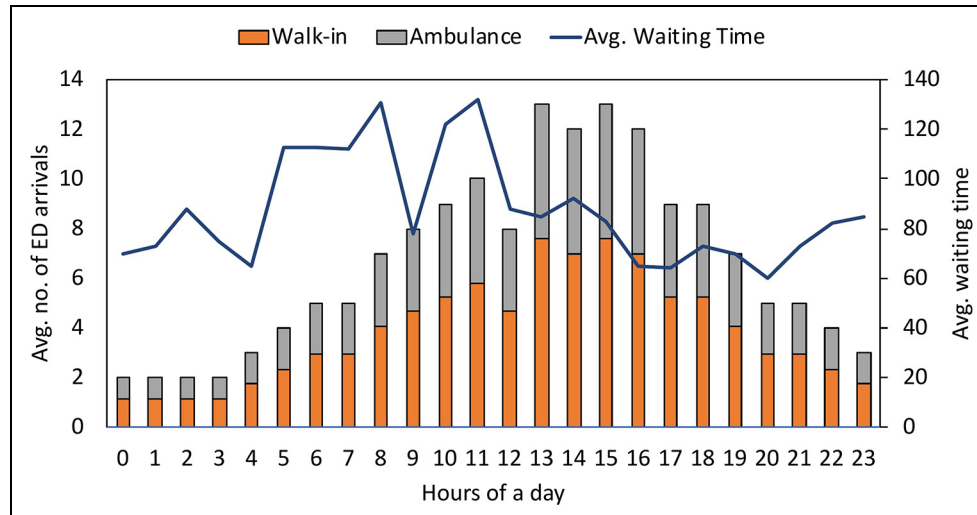


Figure 2. Hourly arrival rates by the patients in the ED for a typical day.

Table 2. Theoretical Probability Distribution of the major stages in the ED patient flow.

Stages	Probability Distribution
Registration	Triangular ~ (2, 3, 5)
Triage	Triangular ~ (5, 10, 15)
Nurse Assessment Time	Lognormal ~ (15,10)
Doctor Assessment Time	Lognormal ~ (20,5)
Physician Assistant Assessment Time	Lognormal ~ (15,5)
X-ray Test	Lognormal ~ (16.53, 8.15)
Computed Tomography (CT) Test	Normal ~ (16.9, 6.84) and lies within (5, 30)
Interventional Radiology (IR) Test	Normal ~ (78.1, 28.7) and lies within (15, 150)

time. The goodness-of-fit test of the arrival time data confirmed Poisson distribution with time-varying mean to be an appropriate distribution for the number of patients arriving at the ED (p -value = 0.62). The registration and triage processing time data collected from the time study did not fit any theoretical distribution. Therefore, a triangular distribution was assumed, and its parameters were established by using the time-study data. On the other hand, the probability distribution for all the other stages was determined based on the goodness-of-fit test. Table 2 shows the distribution of the key stages in ED patient flow.

Note that the data pertaining to laboratory tests (CT and IR) fitted well to a normal distribution, but may yield negative or implausible service time values. To avoid such unrealistic cases, it was modeled as a truncated normal distribution by bounding the sampled values from both sides. Historical data and expert opinion were used to ensure appropriate values for these bounds.

4.3. Modeling and evaluation of current process

4.3.1. Discrete-event simulation model development. The simulation model of the ED was designed and developed using Simio® simulation software by adopting the process and sequence described in Section 3.1. As the system under study involves complex human interactions and decisions, it would not be possible to replicate it entirely in the simulation model. Therefore, the following assumptions were made based on the interaction with the hospital staff and engineers.

- Waiting room capacity is sufficiently large to accommodate the incoming demand.
- Each type of radiology test will cover multiple tests in a single flow.
- Administrative process work of billing and other paperwork is not considered; patients are assumed to exit the system once the patient is discharged from the ED.

Table 3. Validation of key measures in the model.

Validation Measure	Actual Data	Simulation Data	p-value
Average arrivals per day	82	78	0.023
• Ambulance	34	31	0.048
• Walk-in	48	47	0.021
Acuity level distribution			
• Level 1	41.96%	39.4%	0.037
• Level 2	11.8%	12.2%	0.012
• Level 3	35%	37.3%	0.034
• Level 4	11.1%	11.02%	0.007
• Level 5	0.03%	.08%	0.021
Average waiting time	87 min	89.9 min	0.036

As the ED operates for 24 hours, it is modeled as a non-terminating system. To eliminate the initial bias and allow the model to attain steady state, a warm-up length is determined using Welch's method.⁴⁹ Based on several pilot runs of the DES model, it was observed that the average value of performance measures stabilized by 30 days. Therefore, this time period is used as the warm-up length. Besides, a run length of 365 days is chosen based on empirical analysis as it is sufficiently long to allow all types of ED events (including rare occurrences such as walk-ins with level 1 severity). To estimate the required number of replications, the DES model is initially run for 10 replications (n_0) resulting in an average patient waiting time of 92.3 minutes and a standard deviation of 15.2 minutes. Given these measures, the half-width of 95% confidence interval is $t_{n_0-1, 1-\frac{0.05}{2}} \frac{s}{\sqrt{n_0}} = 2.23 \frac{15.2}{\sqrt{10}} = 10.72$. The required number of replications (n) for a specific error margin (h) can be approximated as $z_{1-\frac{0.05}{2}}^2 \frac{s^2}{h^2}$.⁵⁰ In this study, a smaller error margin of 3 minutes is specified, and thus the simulation model is run for 50 replications ($n \sim 1.96^2 \frac{10.72^2}{3^2}$).

4.3.2. Model verification and validation. The ED under study has complex flows and process, which primarily depend on the patient's acuity level. The verification efforts included different techniques suggested by Law.⁵⁰ First, the model was checked by subject matter experts to ensure accurate translation of the conceptual model into a computerized representation. Second, the debugging tools in Simio[®] were used to verify the logical flow in the system. Finally, different input parameters settings were used to run the simulation model, and the corresponding model outputs were examined for reasonableness. To ensure high face validity and substantiate the model assumptions, the process operations manager at the ED, who is also familiar with simulation modeling, was involved during model conceptualization/development. The model is also validated by comparing the key measures from the simulation model with the actual values observed in the real system.

Table 4. Performance measure for the current ED setting.

Performance Measure	Average Value (Standard Deviation)
Nurse Utilization	79.2% (1.2%)
Doctor Utilization	86.4% (0.8%)
Physician Assistant Utilization	12.1% (1.2%)
Trauma Bed Utilization	37.3% (1.2%)
General Bed Utilization	58.4% (1.3%)
Waiting Time (in minutes)	89.9 (2.9)

Table 3 presents a comparison of the three validation measures (number of arrivals, acuity level distribution, and average waiting time). A t -test of the measures confirmed no significant difference between the current system and the simulation model, thereby validating the model.

4.3.3. Process performance measures. One of the critical measures considered by patients as well as hospital stakeholders to evaluate an ED is the waiting time or door-to-provider time (i.e., the time a patient waits in the ED before being assessed by a medical doctor or physician assistant). An ED experiencing low waiting time is expected to have less overcrowding, thereby leading to higher service quality. Another vital measure considered by any hospital administration is the utilization of resources (nurses, physician assistants, doctors, beds). Typically, a hospital expects to efficiently utilize all the resources without overburdening them. Therefore, in this research, these two types of measures, namely, waiting time for doctor and utilization of resources, are considered to evaluate the ED process performance.

4.3.4. Results of the case study. The performance measure obtained from the simulation model for the current ED setting (baseline model) is illustrated in Table 4. It can be observed that the utilization of human resources is imbalanced. The nurses and doctors are constantly overworked

as they spend the majority of their time (79.2% and 86.4%, respectively) with patients. On the other hand, the physician assistants are poorly utilized as they spend less than 15% of their time in serving the patients. The main reason for this discrepancy is that more than 80% of the arriving patients have an acuity level of 1, 2, or 3, and therefore require assessment/treatment by a doctor. As a result, the doctor is always busy, which in turn leads to high patient waiting time (~90 minutes).

5. Design and evaluation of alternative configurations

5.1. Proposed alternative scenarios

In this section, five different alternative ED settings are considered. The first scenario evaluates the hospital's plan for ED expansion, whereas the last four scenarios are proposed based on the critical inefficiencies highlighted by the baseline model—imbalanced utilization of resources, high waiting time, and overburdened doctors.

- **Scenario 1: Impact of Hospital's Proposed Plan of Capacity Expansion**
- Hospital management realized the need for improving the system performance and initiated the effort to establish an alternative setting by obtaining inputs from different stakeholders—frontline staff, facility manager, administrators, and senior leadership. Therefore, the first scenario represents the hospital's tentative plan for expansion, which focuses on increasing the total number of beds from 27 to 52 and hiring an additional nurse.
- **Scenario 2: Impact of Allowing Physician Assistants to Serve Level 3 Cases**
- The analysis of EMR data showed that 55% of the patients coming to the ED were classified as level 3 acuity. Under the current process setting, the doctors are solely responsible for the assessment of these patients, which then leads to an imbalance in utilization among doctors and physician assistants. Therefore, to overcome such a situation, Scenario 2 evaluates the impact of evenly distributing the workload of level 3 patients among doctors and physician assistants. As the physician assistants are skilled and trained at treating most level 3 cases, this alternative is feasible in practice.
- **Scenario 3: Impact of Capacity Expansion (hiring an additional doctor)**
- During the time study, it was observed that the patients were primarily waiting for doctors compared with other resources such as beds and nurses. This was also confirmed based on the analysis of the baseline model, where the waiting time for a doctor was observed to be a major bottleneck

during the first shift (8 am–8 pm). Therefore, Scenario 3 evaluates the impact of expanding the resource capacity by employing another doctor during the first shift.

- **Scenario 4: Impact of Capacity Expansion with Process Change**
- This scenario evaluates the combined impact of a process change (Scenario 2) and capacity expansion (Scenario 3). In other words, Scenario 4 considers the impact of adding another doctor as well as evenly distributing the level 3 cases among doctors and physician assistants.
- **Scenario 5: Impact of Integrating Process Change and Optimizing Resource Allocation**
- The ED under study has a constant number of nurses, doctors, and physician assistants scheduled throughout the day. However, optimizing the workforce level based on historical demand patterns could improve system performance. Therefore, this scenario incorporates the process change (Scenario 3) and then optimizes the number of health professionals required for each shift with the objective of minimizing patient waiting time, while ensuring a certain level of resource utilization. The scenario is developed and analyzed using the OptQuest add-in in Simio, which includes a collection of search procedures and optimization solvers.⁵¹ Moreover, the utilization of each resource is constrained to be between 70% and 80% as it is the recommended level for ensuring a productive workforce with fewer errors.⁵²

5.2. Experimental results for alternative scenarios

Figures 3 and 4 show a comparison of the average waiting time and resource utilization across all the scenarios under consideration. It can be observed that the hospital's tentative plan (Scenario 1) did not yield a substantial decrease in the average patient waiting time. This occurs because Scenario 1 only increases the capacity for resources (trauma and general beds) which do not impede the patient flow through the ED. Recall that the analysis of the baseline model attributed busy doctors as the primary reason for delays and overcrowding. As Scenario 1 does not add capacity at the bottleneck, it also leads to lower utilization of the beds while the utilization of human resources is not drastically affected. On the other hand, it can be observed that the average patient waiting time is lower in Scenarios 2–5 compared with the baseline model as these configurations focus on reducing the workload of the doctors (bottleneck stage). Likewise, scenarios adopting a change in the process of serving level 3 cases (Scenarios 2, 4, and 5) have considerably reduced the burden on the doctors and improved the utilization of the physician assistants compared with the baseline model. Specifically, Scenario 5

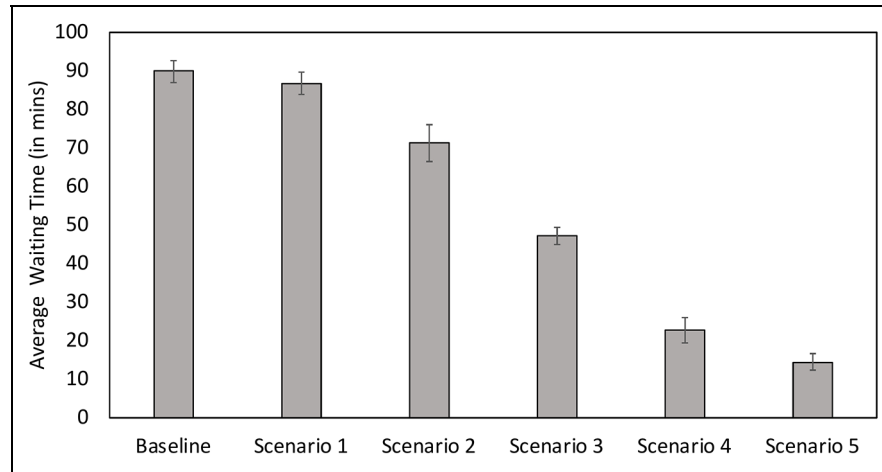


Figure 3. Comparison of the average waiting time across all scenarios.

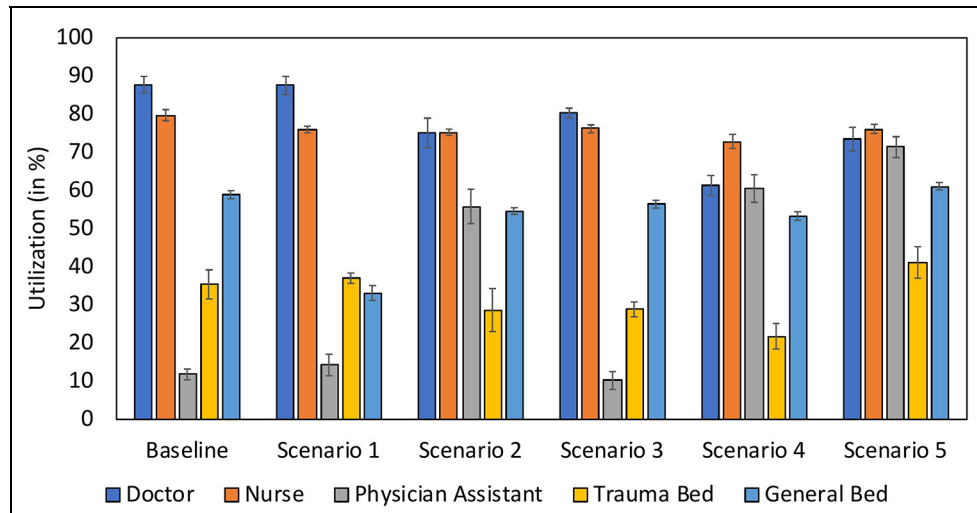


Figure 4. Comparison of resource utilization across all scenarios.

ensured a balanced utilization of human resources (doctors, physician assistants, and nurses) without overburdening them by optimizing the workforce requirement per shift. As shown in Figure 5, the best workforce plan to minimize waiting time and achieve a balanced utilization of human resources is to downsize the nurse and physician by one personnel in Shift 2, and employ two additional physician assistants in Shift 1.

A Tukey multiple comparison test (at $\alpha = 0.05$) is conducted to test the statistical significance of the observed trends in the performance measures across different scenarios. As expected, Scenario 1 (hospital's proposed plan of expansion) did not yield significantly better performance compared with the baseline model in terms of patient wait time and resource utilization (i.e., $p\text{-value} > 0.05$). On the other hand, scenarios dealing with process change yield a

significantly lower waiting time and better utilization of physician assistants as opposed to the current operation.

5.3. Impact of increased ED arrivals

In the initial analysis, the estimates from the EMR data were used to model the patients' arrival as Poisson distribution with a time-varying rate. During the last 5 years, the ED experienced a 25% increase in total patient visits. Moreover, the same trend is expected to continue, and hospital management has projected up to a 30% increase in ED arrivals over the next 10 years. Although management is planning to expand their capacity to accommodate the imminent influx, it is also necessary to identify a robust configuration which could handle the increasing demand in the long run. Therefore, the mean value of the time-

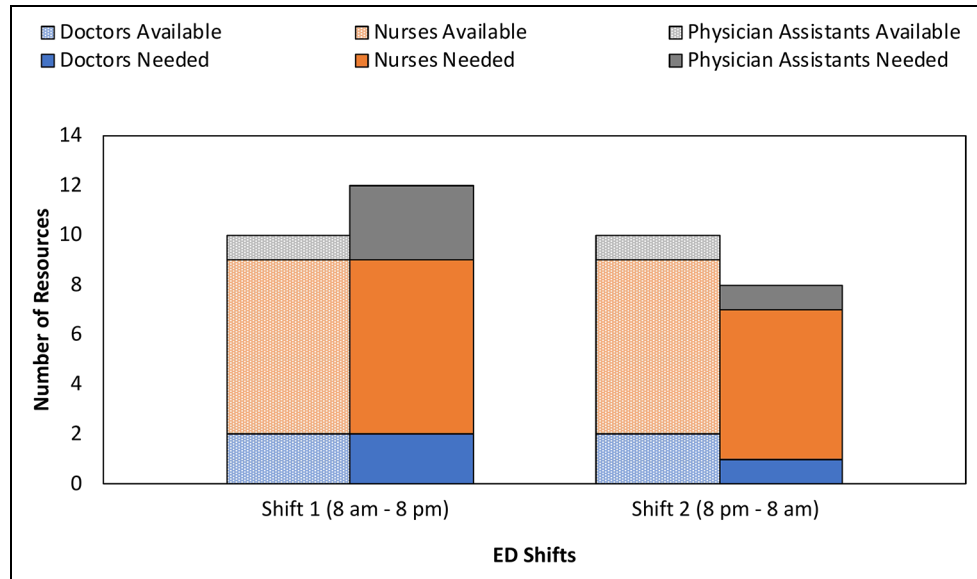


Figure 5. Comparison of resource utilization across all scenarios.

Table 5. Impact of increased arrivals on waiting time and resource utilization.

Proportion Increase in Mean Arrival Rates	Scenarios	Waiting time (in min)	Utilization (%)		
			Doctors	Nurses	Physician Assistants
10%	Baseline	98.1	86.5	81.1	11.3
	Scenario 1	90.2	87.9	82.1	12.7
	Scenario 2	75.3	75.1	80.2	54.3
	Scenario 3	52.5	80.6	78.7	12.2
	Scenario 4	27.5	63.6	76.1	60.1
	Scenario 5	17.2	75.3	77.3	74.6
20%	Baseline	101.3	91.3	87.3	13.6
	Scenario 1	92.8	89.4	84.1	13.1
	Scenario 2	77.6	77.2	82.5	54.4
	Scenario 3	55.9	84.7	82.9	13.0
	Scenario 4	30.5	67.1	81.4	62.7
	Scenario 5	19.1	76.7	78.1	76.5
30%	Baseline	104.7	93.2	88.4	14.2
	Scenario 1	94.3	90.2	85.5	14.0
	Scenario 2	78.8	77.0	86.1	55.0
	Scenario 3	57.9	85.0	84.8	13.1
	Scenario 4	32.4	69.3	87.1	67.0
	Scenario 5	21.5	78.1	78.9	77.1

varying arrival rates is increased by 10%, 20%, and 30%, and its impact on the performance measures is analyzed; all the other parameters remain the same as the initial setting. Table 5 shows a comparison of the performance measures for increased arrival rates. The optimal workforce plan based on Scenario 5 did not change when the arrival rates increased by 10%. However, if the arrival rates increased by 20% or 30%, then it is necessary to have two doctors, seven nurses, and three physician assistants in both the shifts.

Unlike the preliminary analysis, Scenario 1 has significantly lower waiting time than the baseline model, when the arrival rates increase by 10%, 20%, or 30%. However, consistent with the initial findings, Scenario 1 does not significantly improve resource utilization compared with the baseline model if the demand increases (p -value > 0.05). Further, the results also indicate the waiting time under Scenarios 2–5 to be significantly better than the current setting (p -value < 0.05) when the arrival rates increase by up to 30%. As Scenario 5 has adopted both process

change and workforce optimization, it is able to perform better even for future cases as it consistently resulted in the least waiting time for all values of increased arrival rates under study.

6. Discussion on factors influencing the implementation of alternative configuration

In most simulation-based studies, an alternative chosen for implementation is not solely based on its ability to achieve the best improvement in process performance but also its consequences on other factors such as the required capital investment and time commitment. In this section, a discussion and comparison of each scenario based upon the five criteria, namely, cost of adoption, ease of implementation, responsiveness, risk, and performance, are presented.

Cost of Adoption: An analysis pertaining to the expected cost for adopting each alternative is presented in the Appendix. Scenario 5 reduces the operational cost for the current demand pattern as it downsized on expensive resources and increased capacity of less-costly resources. However, Scenario 5 will require substantial capital investment as the demand grows by more than 20%. In the long term, Scenario 2 requires the least cost for implementation as it can be accomplished using existing resources. On the other hand, Scenario 1 (hospital's plan of capacity expansion) requires the highest capital investment followed by Scenarios 3 and 4, as all these alternatives need extra resources even for short-term implementation.

Ease of Implementation: With respect to the ease of implementation, Scenario 1 will impede the current patient flow due to the installation of additional beds and the associated change in layout. In addition, it also requires the recruitment of a nurse, which is difficult due to a skill shortage. Scenario 2 requires a minimal change in the triage process. Even though this might be a setback for the triage nurses upon implementation, they would eventually adapt to the new protocol given their experience. Scenarios 3–5 do not disrupt the current patient flow or operations. However, hiring a healthcare practitioner such as a doctor is time-consuming and challenging due to supply shortage, especially in medically underserved areas. Even though Scenario 5 downsizes physician capacity and requires only two additional physician assistants to address the current needs, there will be a necessity to employ additional doctors and nurses in the near future. Scenario 2 is the easiest alternative to implement, followed by Scenarios 3, 4, and 5, whereas Scenario 1 is the most laborious and time-consuming configuration to implement.

Responsiveness: Because of the projected growth in the aging population, ED visits are also expected to increase in the future. With respect responsiveness (ability to handle increased arrivals), Scenario 1 is only marginally better than the baseline model based on the process performance. Therefore, the increased arrival rate will worsen the patient waiting time. However, Scenarios 2–5 are able to handle the increased arrivals rates without causing substantial inconvenience to the patients. Scenario 5 has the best capability to handle fluctuating demand as it adjusts the capacity accordingly, followed by Scenarios 4, 3, and 2.

Risk: As humans are involved in the ED process, the potential risks and consequences of any change in the system should be considered and assessed with utmost care. Any process change should not affect the quality of care provided by the ED. Among the scenarios evaluated, Scenario 3 has no foreseeable risk as it only increases the resource capacity. Scenario 1 interrupts the existing process flow during the implementation phase and therefore could affect patient experience and employee engagement. Alternatives 2, 4, and 5 possess a relatively higher risk, as complicated level 3 cases could be assigned to physician assistants instead of doctors. In such situations, there will be a delay in treatment begin time as the patient will have to be reassigned to the next available doctor. Although such an occurrence is rare and does not impose any health risk, it could be avoided by establishing a proper triage protocol and effective training of triage nurses. Moreover, Scenario 5 is optimized for the current demand pattern and will be the worst-performing alternative if there a sudden surge in ED visits as a result of threats such as epidemics or terrorist attacks.

Performance: All the proposed scenarios reduce the waiting time of patients in the ED, which is a significant concern in the current system. Scenario 1 reduces the waiting time by a mere 3%, whereas Scenarios 3, 4, and 5 reduce it by 47%, 74%, and 84%, respectively. The utilization of resources improved in Scenarios 1 and 3 but was still imbalanced among resources. However, Scenarios 2 and 4 reduced the burden on the doctor and Scenario 5 efficiently balanced utilization across all resources. In particular, Scenario 5 dominates the other alternatives with respect to process performance measures as it has a maximum reduction in waiting time while balancing resource utilization.

Table 6 ranks the proposed scenarios with respect to the different factors affecting implementation. It is evident that Scenario 1 is always dominated by another alternative and is never ranked first in any of the five criteria considered. Scenario 2 improves service quality (or waiting time) by 20% and requires a nominal investment from the hospital. Moreover, it is very easy to implement as it requires a

Table 6. Ranking of proposed scenarios across different factors.

Alternative Scenarios	Cost of Adoption	Ease of Implementation	Responsiveness	Risk	Performance
Scenario 1	5	5	5	2	5
Scenario 2	1	1	4	3	4
Scenario 3	3	2	3	1	3
Scenario 4	4	3	2	3	2
Scenario 5	2	4	1	5	1

minimal change in the triage process. Therefore, it would be a very attractive solution for implementation in the short term. Scenario 3 has the least risk involved and consistently ranks third for most of the criteria considered, thereby achieving a balance among the factors under consideration. Therefore, if the decision-maker considers all five factors to be equally important, then Scenario 3 would be viable. Scenario 5 achieves the best improvement in performance and is very well suited to handle future demand scenarios. Moreover, its cost of implementation in the long term is competitive compared with Scenarios 3 and 4. Note that Scenario 5 might reduce operational costs in the short term, but downsizing physician workforce for short-term benefits and then expanding it for future demand might become time-intensive and challenging because of physician shortages. Therefore, Scenario 5 is an attractive long-term alternative for the hospital, especially when the demand grows by 20%.

7. Conclusions and future work

ED crowding has been a global crisis over the last few years. In this research, the impact of several process reengineering strategies (process change, workforce planning, and capacity expansion) on the performance of an ED is studied using DES. The proposed alternatives were evaluated with respect to the multiple criteria such as ease of implementation, cost of adoption, responsiveness to demand fluctuation, performance (waiting time and resource utilization), and risk. Based on the simulation analysis of the current operations, two key problems are identified: high patient waiting time and inefficient utilization of available resources (i.e., high utilization of doctors versus low utilization of physician assistants). This is mainly because fewer doctors are responsible for treating the majority of the patients. As a result, different strategies to alleviate this problem are evaluated and compared with the current operations as well as the hospital's proposed expansion plan. It is recommended that the hospital adopt a process change in the near future to curtail the waiting time by 21%. Further, in the long run, it is suggested to embrace the change in process and optimize workforce allocation. As ED throughput is also affected by the efficiency of a co-located inpatient unit, future work could

consider the two units together and optimize the resources required to achieve better service quality.

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Appendix: Cost Analysis of Alternative Scenarios

It is evident from the results presented in Sections 5.2 and 5.3 that Scenario 5 (the combination of process change and workforce planning) yields the best process performance – patient waiting time and resource utilization. However, each of the alternative configurations requires an additional capital investment as well as recurrent expenditure. This section estimates the incremental cost

associated with each scenario. Table A1 presents the cost breakdown of implementing the changes proposed by the hospital (Scenario 1).

Scenario 1 would incur a fixed cost of \$3,880,000 (\$150,000/bed × 25 beds + \$120,000 for the installation and implementation of software + \$10,000 in direct nurse recruitment cost) and an annual recurring cost of \$342,800 (\$80,000 for the licenses and maintenance + \$262,000 in salaries for employing another nurse for two shifts per year). On the other hand, Scenario 2, which uses the existing resources and adopts a minimal process change, incurs only the fixed cost associated with establishing the new triage protocol and training the triage staff. Based on the interaction with the financial manager at the hospital, the cost associated with Scenario 2 was estimated to be \$10,000. Scenario 3 incurs a fixed cost of \$25,000 in direct physician recruitment and an incremental annual cost of \$500,000 for employing an additional doctor in the first shift. As Scenario 4 includes both process change and capacity expansion, it incurs the combined cost of Scenarios 2 and 3, leading to a fixed cost of \$35,000 and an annual cost of \$500,000. Scenario 5 requires a fixed cost of \$15,000 for recruiting two additional physician assistants in the first shift. While these additional resources increase the annual cost by \$438,000, Scenario 5 also reduces the nurse and doctor requirement in Shift 2, resulting in an annual cost reduction of \$635,100 (a reduction of \$131,400 for nurse + \$500,000 for doctor). In summary, optimizing the workforce allocation based on Scenario 5 results in an annual savings of \$197,100 (\$635,100–\$438,000) for the existing demand pattern. Nevertheless, if the demand grows by 20% in the future, then it is necessary to hire two additional physician assistants without downsizing the existing capacity of two doctors, seven nurses, and one physician assistant. Therefore, this would require an additional annual cost of \$438,000 as opposed to the current operating cost.

Table A1. Incremental Cost Calculation for Scenario 1.

Cost Components	Quantity	Fixed Cost	Annual Cost
High dependency emergency bed	25		
Bed frame and mattress		\$1,250,000	-
Cardiac monitor		\$750,000	-
Hemodynamic monitor		\$445,500	-
Respiratory monitor		\$28,500	-
Neurological monitor		\$1,125,000	-
Glucometer		\$28,750	-
Remote patient monitor		\$122,250	-
Software integration and installation		\$120,000	-
License and maintenance		-	\$80,000
Nurse recruitment and salaries for two shifts	1	\$10,000	\$262,800
Total Cost		\$3,880,000	\$342,800

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